INCH-POUND

MIL-O-71152 (AR)
01 November 1993

MILITARY SPECIFICATION

OBTURATING BAND, PLASTIC SPLIT (MOLDED)

This Specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification provides requirements and quality assurance provisions for obturating band, plastic, split (molded) for use on the Cartridge, 60mm, Short Range, Practice, M766.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, NJ 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1395

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STANDARDS

MILTARY

MIL-STD-109 - Quality Assurance Term and Definitions

MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

12953573 - Ring, Obturating

(Copies of drawings, and publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF TESTING AND MATERIAL (ASTM)

ASTM-D256 - Tests for Impact Resistance of Plastics and Electrical Insulating Materials
ASTM-D638 - Test for Tensile Properties of Plastics
ASTM-D648 - Test for Deflection Temperature of Plastics Under Flexural Load
ASTM-D790 - Tests for Flexural Properties of Plastics and Electrical Insulating Materials

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(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. (See contract provisions for additional precedence criteria).

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings, specifications, and this specification.

3.1.1 Certification. Certification must be furnished by the molder for each lot of bands certifying that the plastic used in molding complies with the requirements of 3.1.

3.2 Molded parts. Obturators shall comply with the applicable requirements specified on drawing 12953573 and shall be produced by the injection molding process.

3.3 Translucency. Molded parts shall be sufficiently translucent to inspect for voids.

3.4 Physical property of molded parts. Material used to produce obturators shall meet the properties listed as follows:

Characteristic	Requirements
Izod impact strength (min)	100 Joules/meter (J/M)
Tensile yield strength (min)	100 Megapascal (MPa)
Flexural Modulus (min)	4500 Megapascal (MPa)
Deflection temperature under a load at 1820 kilopascal (KPa)	145 degrees centigrade

3.5 Proving ground test. The mean muzzle velocity of the test rounds (Group 1, with test obturators) shall not differ more than 5 feet/sec from that of the comparison rounds (Group 2, with accepted obturators) when tested in accordance with 4.4.3.2. The variances of the Group 1, S1 and Group 2, S1, shall be as follows (see 4.4.5.2.2):

$$S1 < S2 \quad \text{or} \quad S1/S2 < 2.27$$

3.6 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

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3.7 Workmanship. The parts shall be uniform in color, free of cracks, voids, burn marks, flashing, contamination, and sprues.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General provisions. The requirements of MIL-A-48078 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4)

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the items as required per Table I and shall be from each mold cavity.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078 and 4.4.3.2.

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Ring, Obturating	1 OF 1	See below
			NEXT HIGHER ASSEMBLY: N/A

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Ring, Obturating (Dwg. 12953573)			
	Examination of defects	100	3.2/3.3	4.4.2.1
	Proving ground test	70	3.5	4.4.3.2
	Physical Properties of Material			
	Izod impact strength (min)	5	3.4	4.4.3.1/4.4.5.1.1
	Tensile yield strength (min)	5	3.4	4.4.3.1/4.4.5.1.2
	Flexural modulus (min)	5	3.4	4.4.3.1/4.4.5.1.3
	Deflection temperature under a load at 1820 kilopascal (kpa)	5	3.4	4.4.3.1/4.4.5.1.4

NOTES:

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4.4 Quality conformation inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078 and the following as applicable.

4.4.1.1 Obturator lot. A lot shall consist of obturators produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revisions, same specification and same specification revision. Each inspection lot shall contain: bands manufactured from one lot of resin and molded in one mold on one machine.

4.4.2 Examination and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified herein. When cited herein, attributes sampling inspection shall be conducted in accordance with TABLE II below, using the inspection levels cited in the Classification of Characteristics paragraphs.

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TABLE II. Attributes sampling inspection

LOT SIZE	INSPECTION LEVELS					
	I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	13	5
91 to 150	*	125	32	13	13	5
151 to 280	*	125	32	32	20	8
281 to 500	*	125	32	32	20	8
501 to 1200	*	125	80	50	20	13
1201 to 3200	1250	125	80	50	32	13
3201 to 10000	1250	125	125	50	32	13
10001 to 35000	1250	315	125	80	50	13
35001 to 150000	1250	315	125	80	50	13
150001 to 500000	1250	500	200	125	50	13
500000 and over	1250	500	200	125	50	13

Numbers under inspection levels indicate sample size; asterisks indicate one hundred percent inspection. If sample size exceeds size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, variable sampling or other types of sampling procedures etc. may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (See 6.6). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative(s) is equal to or better than the specified quality conformance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program plan or inspection system, as applicable.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Ring, Obturating	1 OF 2	12953573
			NEXT HIGHER ASSEMBLY 12953564

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Ring gage fails to pass freely over the outside diameter of the ring while placed on cylindrical plug	(a)	3.2	Gage
102	Width when placed on plug	(a)	3.2	Gage
103	Translucency	(a)	3.3	Visual
104	Presence of voids	(a)	3.2	Visual
105	Physical properties of material			
	Izod impact strength (min)	5	3.4	4.4.3.1/4.4.5.1.1
	Tensile yield strength (min)	5	3.4	4.4.3.1/4.4.5.1.2
	Flexural modulus (min)	5	3.4	4.4.3.1/4.4.5.1.3
	Deflection temperature under a load at 1820 kpa	5	3.4	4.4.3.1/4.4.5.1.4
106	When the obturator is placed on a 2.149 inch diameter plug, the unrestrained obturator shall not fall off of the plug when the plug is held in the vertical position	(a)	3.2	Gage

NOTES:

(a) 20 samples from each mold cavity shall be examined for this feature or dimension. The sample of 20 shall include the first two parts from the beginning of each production run and the last two parts from the end of the run. The remaining samples of (at least sixteen) shall be randomly selected during the production run. Any defects found in any of the 20 samples from each die cavity shall be cause for rejection of all ogive rear sections produced from that particular die cavity.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Ring, Obturating	2 OF 2	12953573
			NEXT HIGHER ASSEMBLY 12953564

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Minor				
201	Surface finish improper	V	3.2	Visual
202	Poor workmanship	V	3.7	Visual

NOTES:

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4.4.3 Testing.

4.4.3.1 Physical properties of material. Representative specimens prepared from each lot of plastic material utilized for fabricating the first article and production lots of obturating rings shall be tested in accordance with 4.4.5.1. Five specimen samples shall be subjected to each physical property test. If any sample fails to meet any of the requirements specified in section 3.4 the lot of the plastic material shall be rejected.

4.4.3.2 Proving ground test. Twenty-five test obturators and twenty-five comparison obturators shall be selected for this test (see Table III). The test shall be performed as specified in 4.4.5.2. The lot shall be rejected if one or more groups of the test obturators fail to meet the misfire accept/reject criteria in Table IV.

TABLE III. Ballistic performance

Group	Obturator	Charge	Temp (°F)	Quant
1	Test	Max	70	25
2	STD	Max	70	25
3	Test	Min	145	5
4	Test	Min	-50	5
5	Test	Max	145	5
6	Test	Max	-50	5

Notes:
1. All projectiles shall be from the same lot and +/- 0.05 lbs of its respective in-flight weight.
2. All rounds shall be assembled with acceptable propelling systems from the same lot.
3. All standard obturators shall be used from the same lot.
4. Group 1, 3, 4, 5 and 6 are the test groups.
5. Group 2 is the comparison group.
6. Standard obturators shall be selected from a lot of obturators that were accepted for use.
7. Rounds from group 1 and 2 shall be alternately fired on the same day.

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TABLE IV Performance test plan.

Categories	Defects	Accept/Reject	
Major:	Misfire	0	1
Major:	The mean muzzle velocity of the test group 1 differs from that of the comparison group 2 by 5 ft/sec (see 4.4.5.2.2)		
Major:	The variance of muzzle velocity group 1 (see 4.4.5.2.2)		

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and test prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.4 herein.

4.4.5 Methods of inspection.

4.4.5.1 Physical characteristics. The specimens shall be subjected to test methods as defined in 4.4.5.1.1 through 4.4.5.1.4 to verify compliance with applicable requirements.

4.4.5.1.1 Izod impact strength. Izod impact strength shall be determined in accordance ASTM-Method D256.

4.4.5.1.2 Tensile yield strength. Tensile yield strength shall be determined in accordance with ASTM-Method D638

4.4.5.1.3 Flexural modulus. Flexural modulus shall be determined in accordance with ASTM-Method D790.

4.4.5.1.4 Deflection temperature. Deflection temperature under a load at 1820 kpa shall be determined in accordance with ASTM-Method D648.

4.4.5.2 Proving ground test. The test shall be performed at a Government proving ground in accordance with the applicable USATECOM Acceptance Test Procedure as follows:

All cartridges shall be fired from a standard, 60mm M224 mortar (hard mount) at an elevation of forty-five (45) degrees. The cartridges with the test obturators shall be separated into groups; group 1 of 25 rounds conditioned for 24 hours at +70 degrees F $\pm$ 2.5 degrees F, group 2 of 25 rounds with the standard obturator

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shall be conditioned at +70 degrees F ± 2.5 degrees F, group 3 of 5 rounds conditioned at +145 degrees F ± 2.5 degrees F, group 4 of 5 rounds conditioned for 24 hours at -50 degrees F ± 2.5 degrees F, group 5 of 5 rounds conditioned at +145 degrees F ± 2.5 degrees F and group 6 of 5 rounds conditioned at -50 degrees F ± 2.5 degrees F. The individual velocities shall be recorded on the firing record. The arithmetic averages shall be calculated and recorded. Temperature of the day, weather and the time of firing shall be recorded. Smear camera shall be used to photograph all samples. The camera shall be positioned to provide photo coverage of the test rounds at 20 feet from the muzzle of the mortar tube.

4.4.5.2.1 Test validity. If for any reason the Government considers that the test conditions have detrimentally affected the test results, the test results shall be declared invalid and a new test shall be performed on additional sample cartridges.

4.4.5.2.2 Statistical test. Calculate S_1 for test group 1 and S_2 for comparison group 2 (see Table III).

Where S_1 = Standard deviation of the muzzle velocity of the twenty-five test samples.

S_2 = Standard deviation of the muzzle velocity of the twenty-five comparison samples.

$\overline{X}_1$ = Average of the muzzle velocities of the twenty-five test samples.

$\overline{X}_2$ = Average of the muzzle velocities of the twenty-five comparison samples.

If $S_1^2 > S_2^2$, Calculate $\frac{S_1^2}{S_2^2} = K$ (see Note 1 and 2)

Note 1: If $K < 2.27$ and $\overline{X}_1$ is within ±5 ft/sec of $\overline{X}_2$, the test lot shall be accepted.

Note 2: If $K > 2.27$, the test shall be rejected.

Note 3: If $S_1^2 < S_2^2$ and $\overline{X}_1$ is within ±5 ft/sec of $\overline{X}_2$, the test lot shall be accepted.

5. PACKAGING

5.1 Packaging, packing, and marking. Packaging, packing, and marking shall be in accordance with the requirements of Level C of MIL-STD-1169.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The components covered by this specification are intended for assembly to M766 Practice Cartridges for use in the 60mm Mortar.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.2).

c. Requirements for submission of first article sample.

6.3 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAT-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Submission of test data. One copy of proving ground ballistic acceptance test data (i.e., firing record) shall be forwarded to each of the following: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: SMCAR-QAT-B, Bldg 62S, Picatinny Arsenal, NJ 07806-5000.

6.5 Submission of alternative quality conformance provisions. All contractor proposed alternative quality conformance provisions will be submitted to the Government for evaluation/approval as directed by the contracting activity.

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6.6 Subject term (key word) listing.

Deflection temperature
Flexural modulus
Izod impact
Tensile strength

Custodian
Army-AR

Preparing activity:
Army-AR

(Project 1395-A267)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-O-71152 (AR)	931101

3. DOCUMENT TITLE: OBTURATING BAND, PLASTIC SPLIT (MOLDED)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial: 201-724-6675 (2) AUTOVON: DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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